

Graphing Quadratic Expressions (Part II)

Standard Form	Intercept Form	Vertex Form
$y = ax^2 + bx + c$	$y = a(x-m)(x-n)$	$y = a(x-h)^2 + k$

Finding the Vertex in Standard Form

If a quadratic expression is written in the form $y = ax^2 + bx + c$, then the x-value of the vertex is $-\frac{b}{2a}$.

Finding the Vertex in Intercept Form

If a quadratic expression is written in the form $y = a(x-m)(x-n)$, where m and n are real numbers, then the x-value of the vertex is $\frac{m+n}{2}$.

* Intercept form is not commonly used to graph a quadratic expression if the corresponding graph does not move through the x-axis.

Finding the Vertex in Vertex Form

If a quadratic expression is written in the form $y = a(x-h)^2 + k$, then (h,k) is the vertex.

Finding the Y-intercept in Standard Form

If $y = ax^2 + bx + c$ is graphed on an xy-plane, then (0,c) is the y-intercept of the graph.

Graph each quadratic expression. Show the coordinates of the vertex, y-intercept, and x-intercepts (if there are any). Draw the line of symmetry (as a dotted line) and write its equation.

$$① \quad 3x^2 - 4x + 1 = y$$

$$a = 3, b = -4, c = 1$$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{(-4)}{2(3)} = \frac{2}{3}$$

$$V_y = 3\left(\frac{2}{3}\right)^2 - 4\left(\frac{2}{3}\right) + 1$$

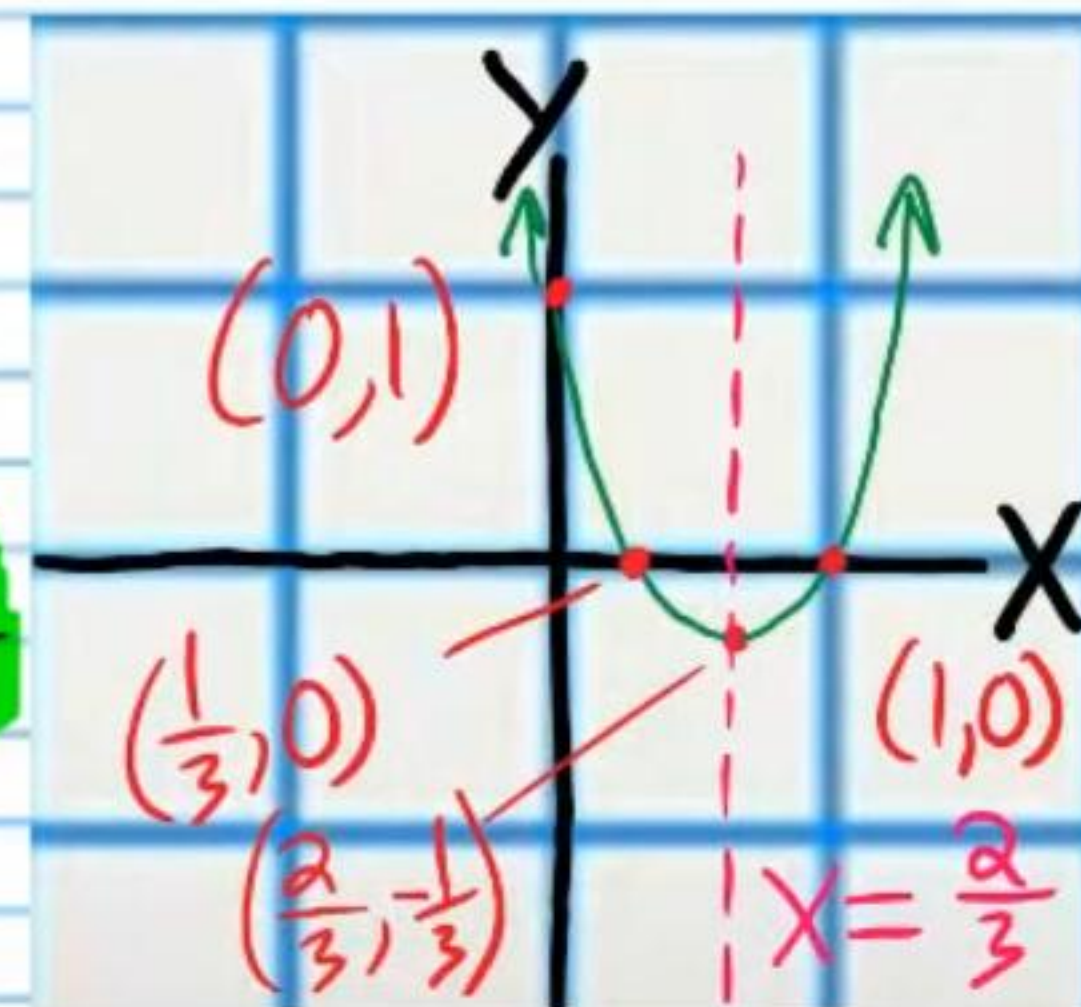
$$V_y = \frac{4}{3} - \frac{8}{3} + 1 = -\frac{1}{3}$$

$$V\left(\frac{2}{3}, -\frac{1}{3}\right)$$

II) Y-Intercept: $x=0$

$$(0,1)$$

III) X-Intercepts: $y=0$



$$3x^2 - 4x + 1 = 0$$

$$(3x-1)(x-1) = 0$$

$$3x-1=0 \text{ or } x-1=0$$

$$x = \frac{1}{3} \text{ or } x = 1$$

$$\left(\frac{1}{3}, 0\right) \text{ \& } (1, 0)$$

$$\textcircled{2} y = -2x^2 + 7x + 4$$

$$a = -2, b = 7, c = 4$$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{7}{2(-2)} = \frac{7}{4} = 1\frac{3}{4}$$

$$V_y = -2\left(\frac{7}{4}\right)^2 + 7\left(\frac{7}{4}\right) + 4$$

$$V_y = -2\left(\frac{7}{4}\right)\left(\frac{7}{4}\right) + 7\left(\frac{7}{4}\right) + 4$$

$$V_y = -2\left(\frac{49}{16}\right) + \frac{49}{4} + 4$$

$$V_y = -\frac{98}{16} + \frac{196}{16} + 4$$

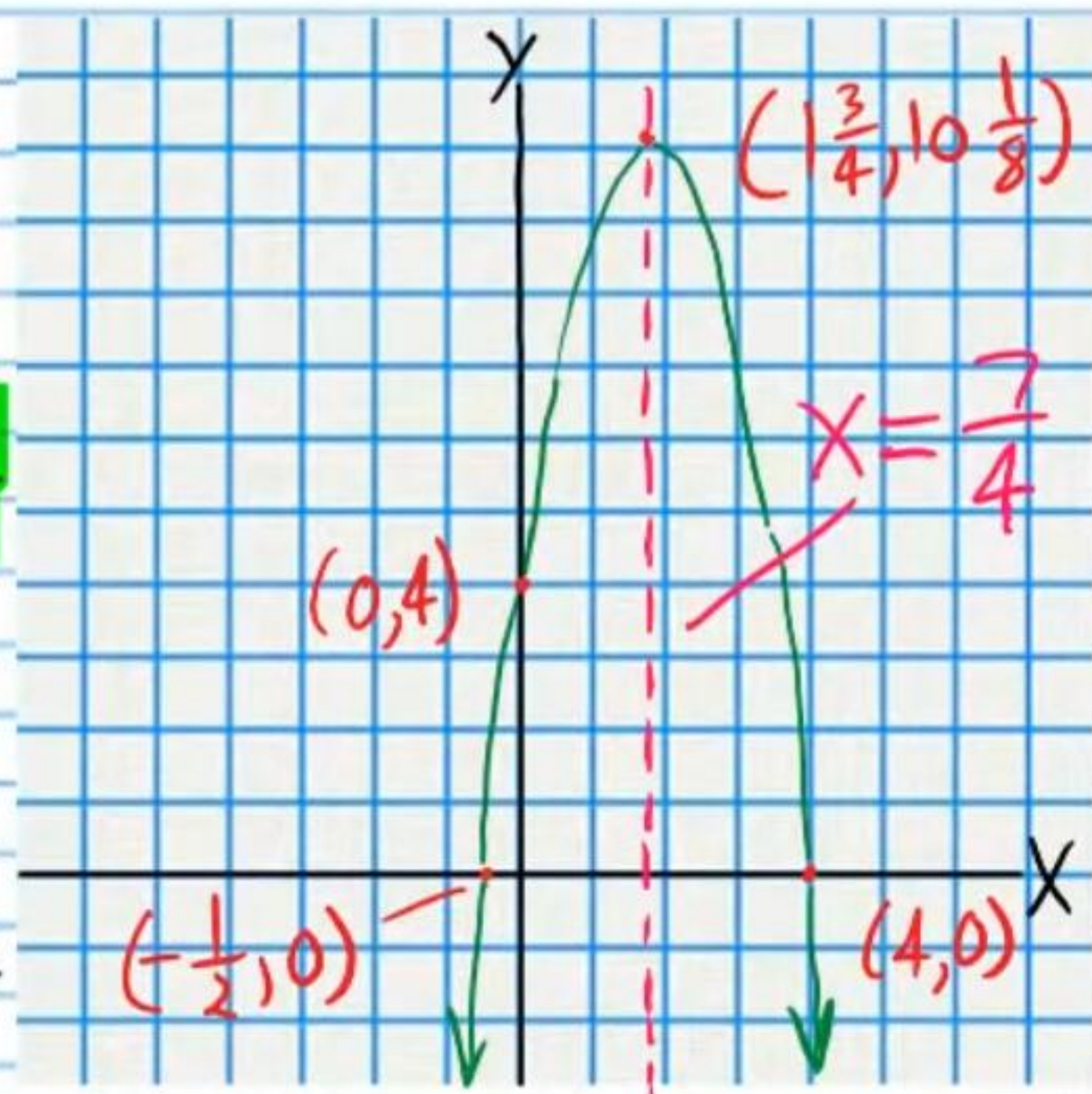
$$V_y = \frac{98}{16} + 4$$

$$V_y = 6\frac{2}{16} + 4$$

$$V_y = 10\frac{2}{16}$$

$$V_y = 10\frac{1}{8}$$

$$V\left(1\frac{3}{4}, 10\frac{1}{8}\right)$$



II) Y-Intercept: $x=0$

$$(0, 4)$$

III) X-Intercepts: $y=0$

$$0 = -2x^2 + 7x + 4$$

$$0 = -1(2x^2 - 7x - 4)$$

$$0 = -1(2x+1)(x-4)$$

$$2x+1=0 \text{ or } x-4=0$$

$$x = -\frac{1}{2} \text{ or } x = 4$$

$$\left(-\frac{1}{2}, 0\right) \text{ \& } (4, 0)$$

$$\textcircled{3} 3x^2 - 8x + 4 = y$$

$$a = 3, b = -8, c = 4$$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{(-8)}{2(3)} = \frac{4}{3} = 1\frac{1}{3}$$

$$V_y = 3\left(\frac{4}{3}\right)^2 - 8\left(\frac{4}{3}\right) + 4$$

$$V_y = \frac{16}{3} - \frac{32}{3} + 4$$

$$V_y = -\frac{16}{3} + 4$$

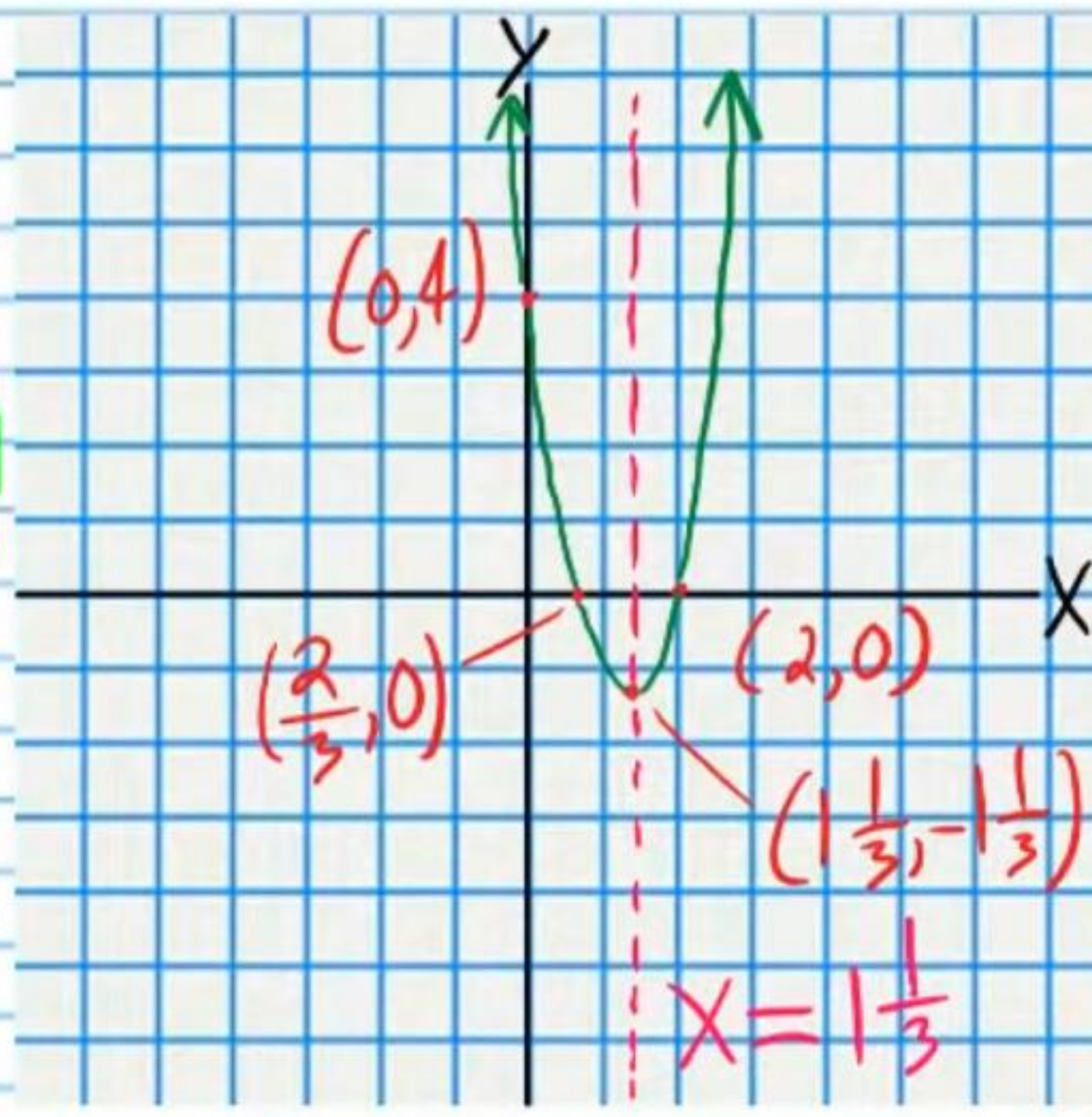
$$V_y = -5\frac{1}{3} + 4$$

$$V_y = -1\frac{1}{3}$$

$$V\left(1\frac{1}{3}, -1\frac{1}{3}\right)$$

II) Y-Intercept: $x=0$

$$(0, 4)$$



III) X-Intercepts: $y=0$

$$3x^2 - 8x + 4 = 0$$

$$(3x-2)(x-2) = 0$$

$$3x-2=0 \text{ or } x-2=0$$

$$x = \frac{2}{3} \text{ or } x = 2$$

$$\left(\frac{2}{3}, 0\right) \text{ \& } (2, 0)$$

④ $y = -4x^2 - 7x + 2$

$a = -4, b = -7, c = 2$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{(-7)}{2(-4)} = -\frac{7}{8}$$

$$V_y = -4\left(-\frac{7}{8}\right)^2 - 7\left(-\frac{7}{8}\right) + 2$$

$$V_y = -\frac{49}{16} + \frac{49}{8} + 2$$

$$V_y = -\frac{49}{16} + \frac{98}{16} + 2$$

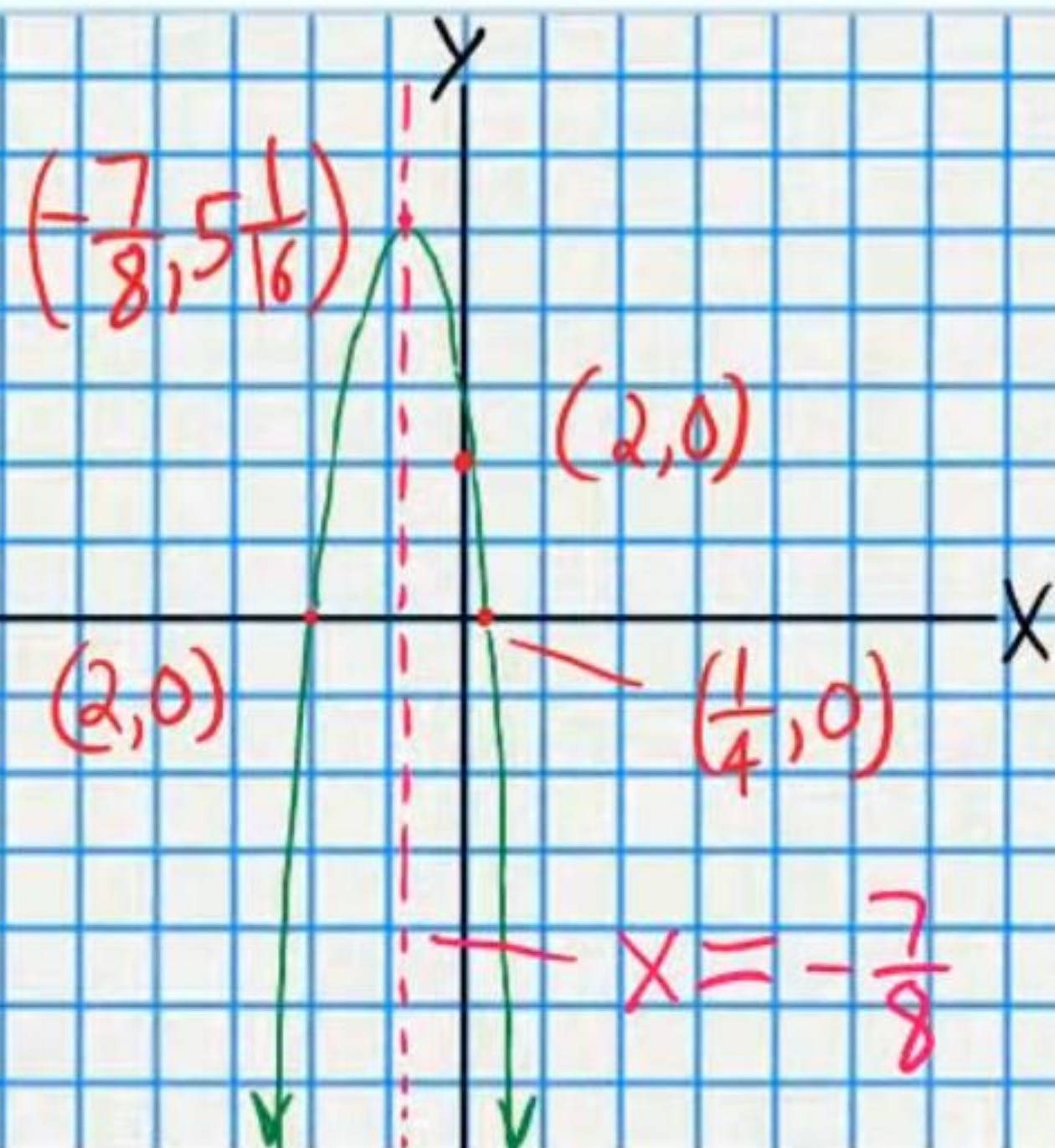
$$V_y = \frac{49}{16} + 2$$

$$V_y = 3\frac{1}{16} + 2$$

$$V_y = 5\frac{1}{16}$$

$$V\left(-\frac{7}{8}, 5\frac{1}{16}\right)$$

II) Y-Intercept: $x = 0$
 $(0, 2)$



III) X-Intercepts: $y = 0$

$$-4x^2 - 7x + 2 = 0$$

$$-1(4x^2 + 7x - 2) = 0$$

$$-1(4x - 1)(x + 2) = 0$$

$$4x - 1 = 0 \text{ or } x + 2 = 0$$

$$x = \frac{1}{4} \text{ or } x = -2$$

$$\left(\frac{1}{4}, 0\right) \text{ \& } (-2, 0)$$

⑤ $y = -x^2 + 3x + 2$

$a = -1, b = 3, c = 2$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{3}{2(-1)} = \frac{3}{2}$$

$$V_y = -\left(\frac{3}{2}\right)^2 + 3\left(\frac{3}{2}\right) + 2$$

$$V_y = -\frac{9}{4} + \frac{9}{2} + 2$$

$$V_y = -\frac{9}{4} + \frac{18}{4} + 2$$

$$V_y = \frac{9}{4} + 2$$

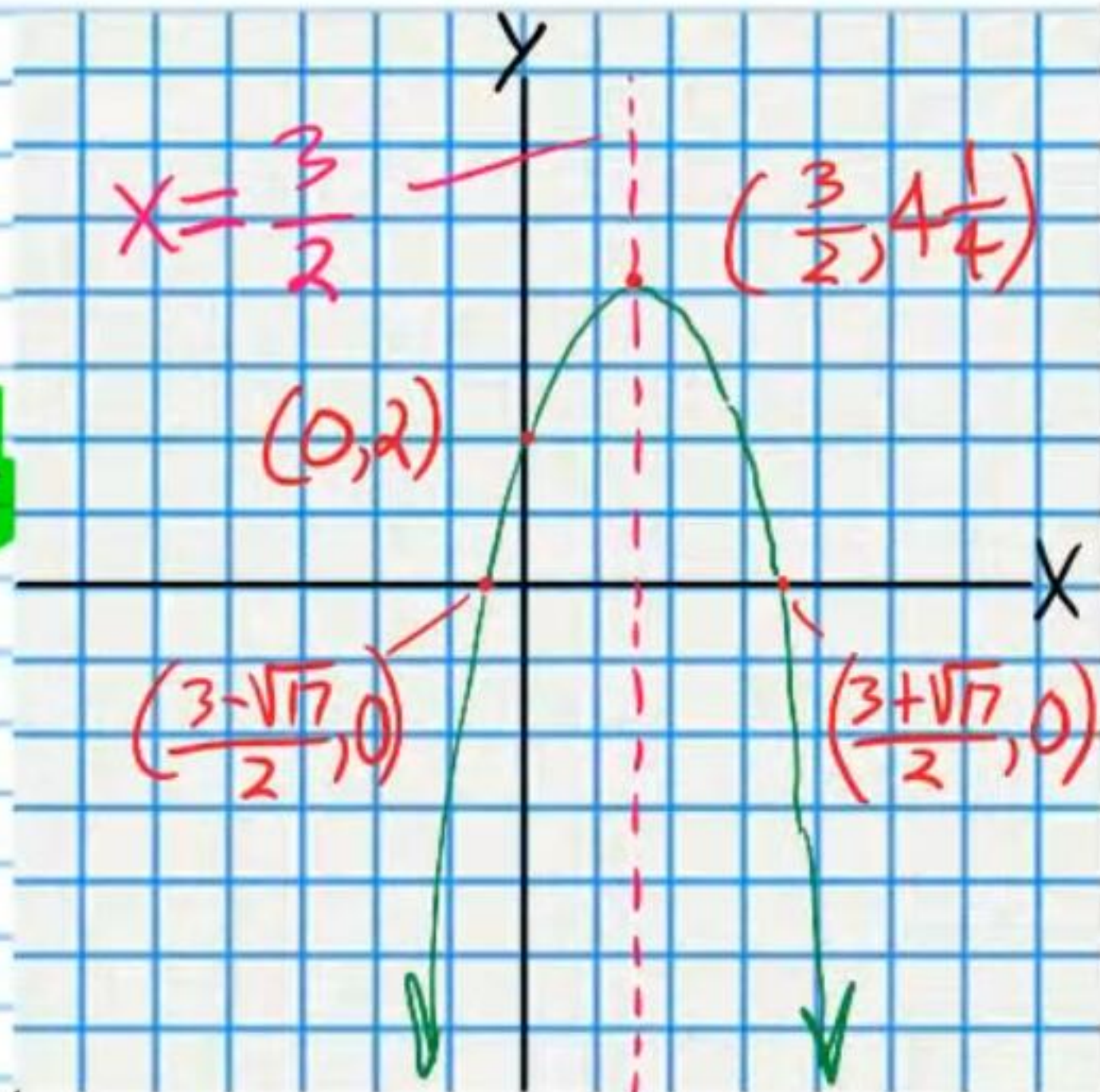
$$V_y = 2\frac{1}{4} + 2 = 4\frac{1}{4}$$

$$V\left(\frac{3}{2}, 4\frac{1}{4}\right)$$

II) Y-Intercepts: $x = 0$
 $(0, 2)$

III) X-Intercepts: $y = 0$

$$0 = -x^2 + 3x + 2$$



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(3) \pm \sqrt{3^2 - 4(-1)(2)}}{2(-1)}$$

$$x = \frac{-3 \pm \sqrt{9 + 8}}{-2}$$

$$x = \frac{-3 \pm \sqrt{17}}{-2}$$

$$x \approx -0.56 \text{ or } x \approx 3.56$$

$$\left(\frac{3 - \sqrt{17}}{2}, 0\right) \text{ \& } \left(\frac{3 + \sqrt{17}}{2}, 0\right)$$

$$⑥ \quad 2x^2 + 4x + 1 = y$$

$$a=2, b=4, c=1$$

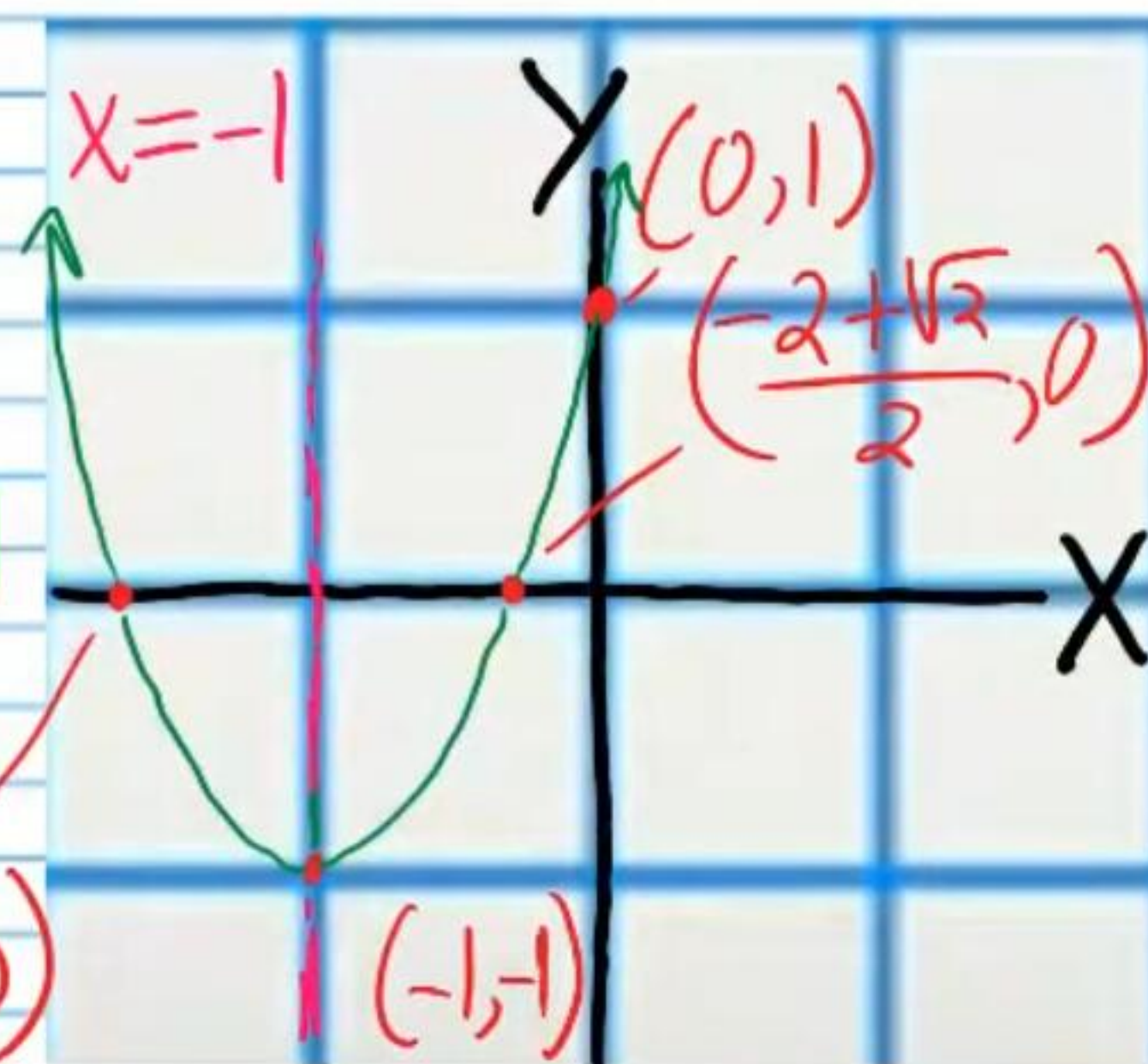
I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{4}{2(2)} = -1$$

$$V_y = 2(-1)^2 + 4(-1) + 1 = -1$$

$$V(-1, -1)$$

$$\left(\frac{-2-\sqrt{2}}{2}, 0\right)$$



II) Y-Intercept: $x=0$

$$(0, 1)$$

III) X-Intercepts: $y=0$

$$2x^2 + 4x + 1 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-4 \pm \sqrt{4^2 - 4(2)(1)}}{2(2)}$$

$$x = \frac{-4 \pm \sqrt{16 - 8}}{4}$$

$$x = \frac{-4 \pm \sqrt{8}}{4}$$

$$x = \frac{-4 \pm 2\sqrt{2}}{4}$$

$$x = \frac{-2 \pm \sqrt{2}}{2}$$

$$x \approx -0.292 \text{ or } x \approx -1.7$$

$$\left(\frac{-2+\sqrt{2}}{2}, 0\right) \text{ and } \left(\frac{-2-\sqrt{2}}{2}, 0\right)$$

$$⑦ \quad y = 3x^2 - 5x - 3$$

$$a=3, b=-5, c=-3$$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{(-5)}{2(3)} = \frac{5}{6}$$

$$V_y = 3\left(\frac{5}{6}\right)^2 - 5\left(\frac{5}{6}\right) - 3$$

$$V_y = \frac{25}{12} - \frac{25}{6} - 3$$

$$V_y = \frac{25}{12} - \frac{50}{12} - 3$$

$$V_y = -\frac{25}{12} - 3$$

$$V_y = -2\frac{1}{12} - 3$$

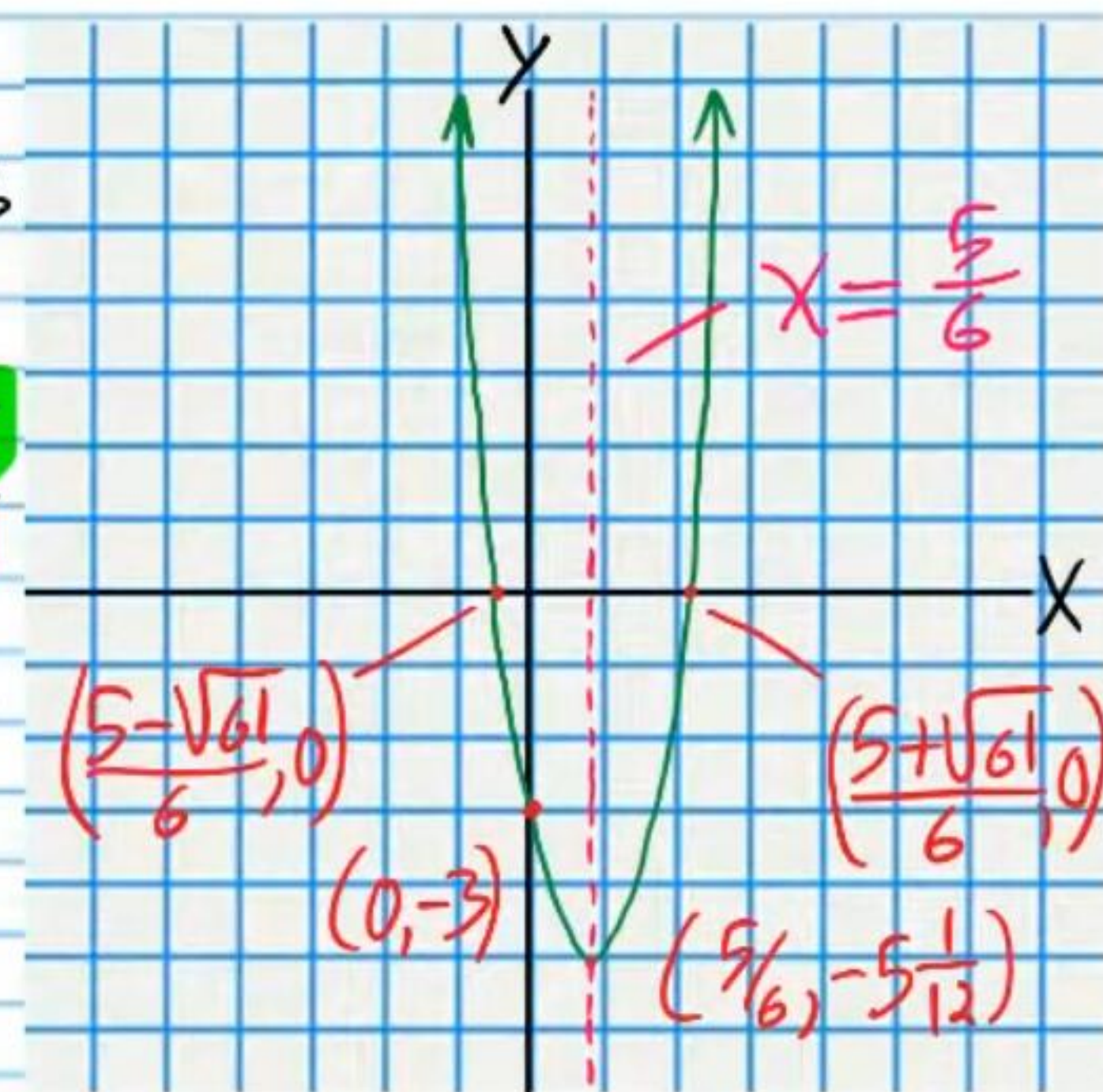
$$V_y = -5\frac{1}{12}$$

$$V\left(\frac{5}{6}, -5\frac{1}{12}\right)$$

II) Y-Intercept: $x=0$

$$(0, -3)$$

III) X-Intercepts: $y=0$



$$0 = 3x^2 - 5x - 3$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(3)(-3)}}{2(3)}$$

$$x = \frac{5 \pm \sqrt{25 + 36}}{6}$$

$$x = \frac{5 \pm \sqrt{61}}{6}$$

$$x \approx 2.13 \text{ or } x \approx -0.468$$

$$\left(\frac{5+\sqrt{61}}{6}, 0\right) \text{ and } \left(\frac{5-\sqrt{61}}{6}, 0\right)$$

$$\textcircled{8} -x^2 - 2x + 5 = y$$

$$a = -1, b = -2, c = 5$$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{(-2)}{2(-1)} = -1$$

$$V_y = -(-1)^2 - 2(-1) + 5$$

$$V_y = 6$$

$$V(-1, 6)$$

II) Y-Intercept: $x = 0$

$$(0, 5)$$

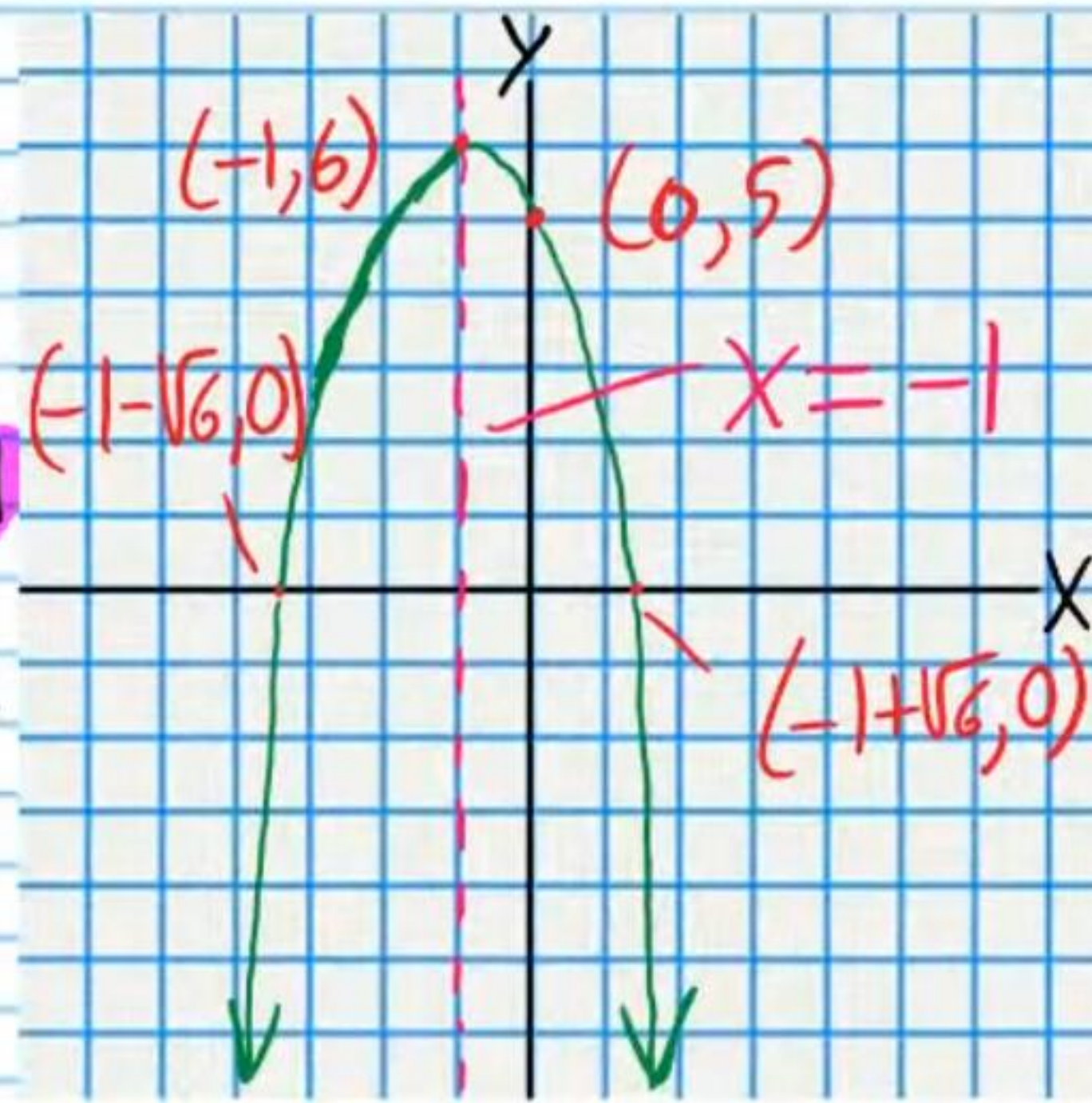
III) X-Intercepts: $y = 0$

$$-x^2 - 2x + 5 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(-1)(5)}}{2(-1)}$$

$$x = \frac{2 \pm \sqrt{4 + 20}}{-2}$$



$$x = \frac{2 \pm \sqrt{24}}{-2}$$

$$x = \frac{2 \pm 2\sqrt{6}}{-2}$$

$$x = \frac{1 \pm \sqrt{6}}{-1}$$

$$x = -1 \pm \sqrt{6}$$

$$x \approx 1.44 \text{ or } x \approx -3.44$$

$$(-1 + \sqrt{6}, 0) \text{ and } (-1 - \sqrt{6}, 0)$$

$$\textcircled{9} y = x^2 + 6x + 10$$

$$a = 1, b = 6, c = 10$$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{6}{2(1)} = -3$$

$$V_y = (-3)^2 + 6(-3) + 10$$

$$V_y = 9 - 18 + 10$$

$$V_y = 1$$

$$V(-3, 1)$$

II) Y-Intercept: $x = 0$

$$(0, 10)$$

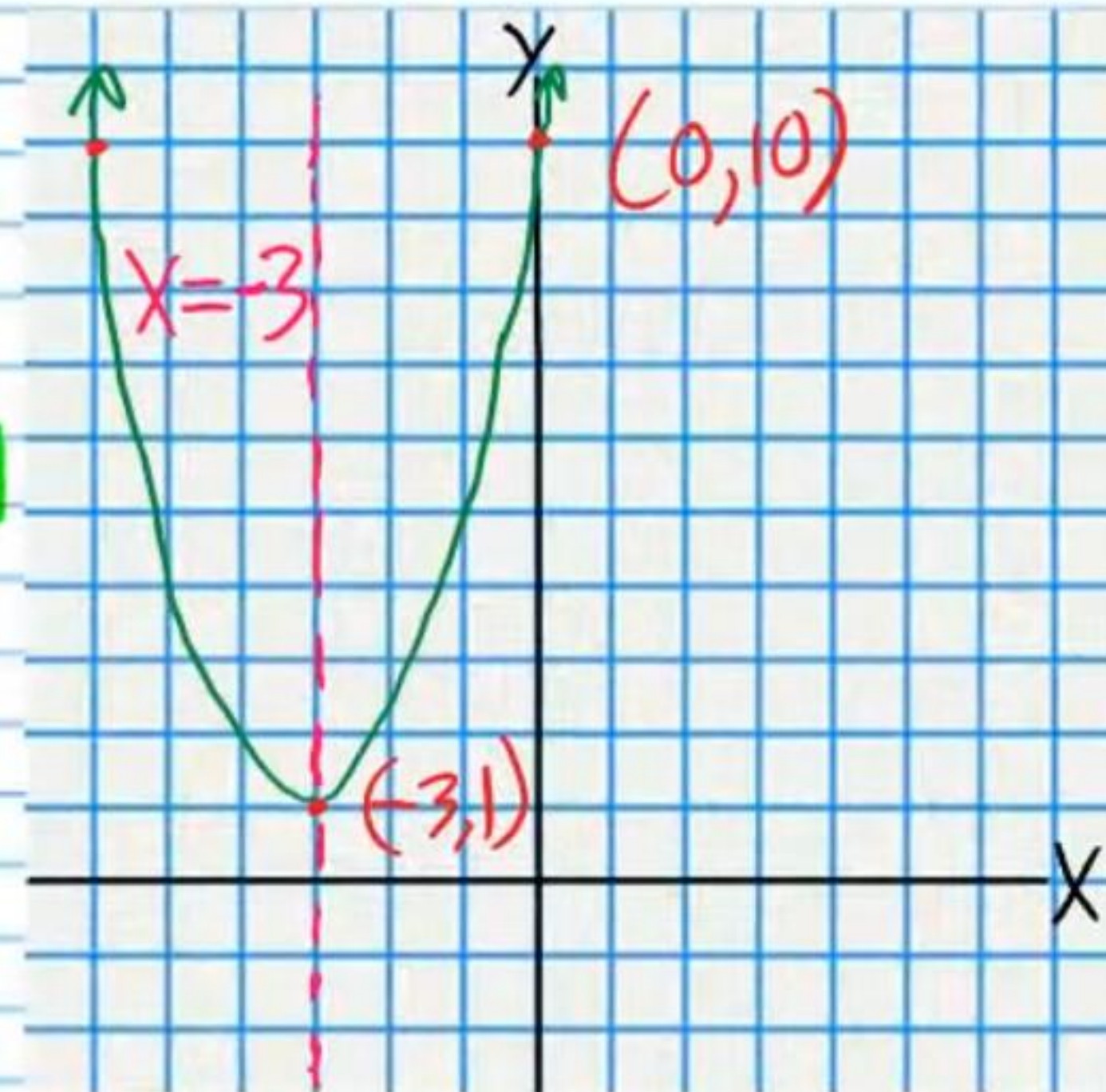
III) X-Intercepts: $y = 0$

$$0 = x^2 + 6x + 10$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-6 \pm \sqrt{6^2 - 4(1)(10)}}{2(1)}$$

$$x = \frac{-6 \pm \sqrt{36 - 40}}{2}$$



$$x = \frac{-6 \pm \sqrt{-4}}{2}$$

STOP

$$x = \frac{-6 \pm 2i}{2}$$

$$x = -3 \pm i$$

$$x = -3 + i \text{ or } x = -3 - i$$

$$\textcircled{10} -2x^2 + 4x - 3 = y$$

$$a = -2, b = 4, c = -3$$

I) Vertex

$$V_x = -\frac{b}{2a} = -\frac{4}{2(-2)} = 1$$

$$V_y = -2(1)^2 + 4(1) - 3$$

$$V_y = -1$$

$$V(1, -1)$$

II) Y-Intercept: $x = 0$

$$(0, -3)$$

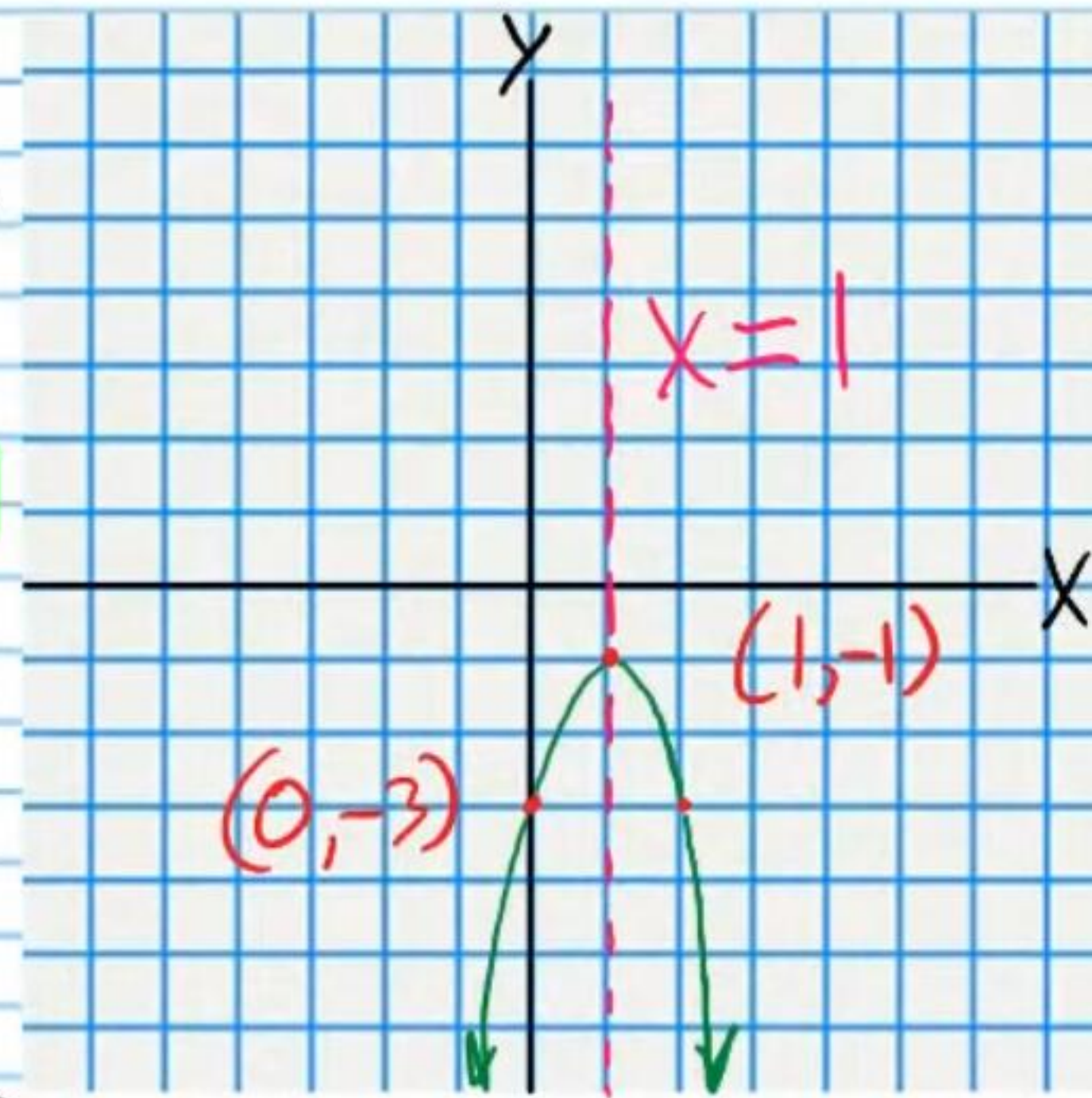
III) X-Intercepts: $y = 0$

$$-2x^2 + 4x - 3 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(4) \pm \sqrt{4^2 - 4(-2)(-3)}}{2(-2)}$$

$$x = \frac{-4 \pm \sqrt{16 - 24}}{-4}$$



$$x = \frac{-4 \pm \sqrt{-8}}{-4}$$

STOP